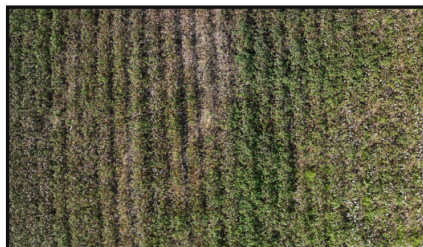


Mask-Guided Cotton Boll Phenotyping Architecture

Pre/post-defoliation UAV imagery → detector-guided lint segmentation → morphology 2.5D review → conservative trait summaries

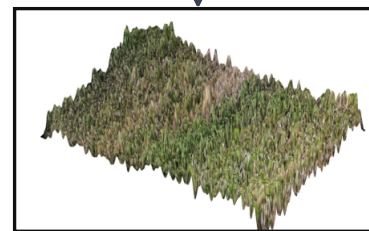
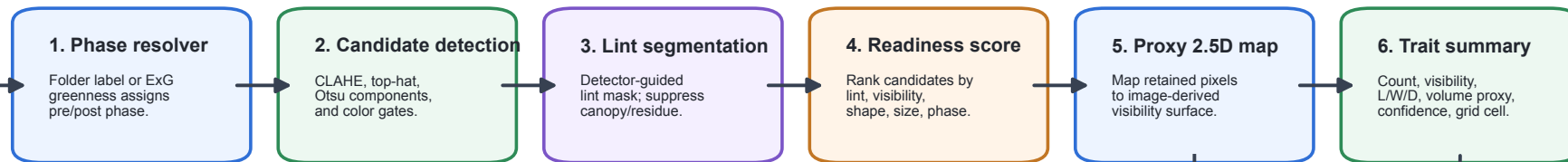
Inputs: aerial visibility intervention
Pre-defoliation RGB



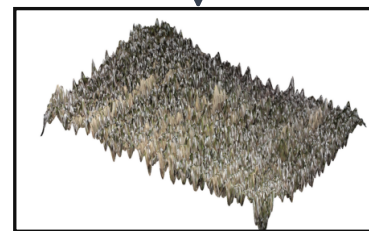
Post-defoliation RGB



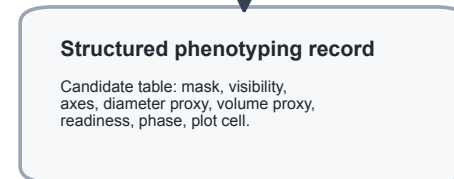
Implemented proxy pipeline



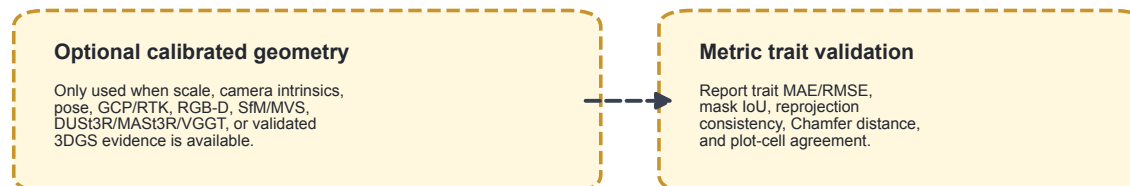
Pre-defoliation proxy 2.5D



Post-defoliation proxy 2.5D



Calibration-dependent branch



Arrow convention

—→ implemented proxy path
- - -→ calibration-only validation